

# System and Network Security

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# Secure Electronic Transaction (SET)

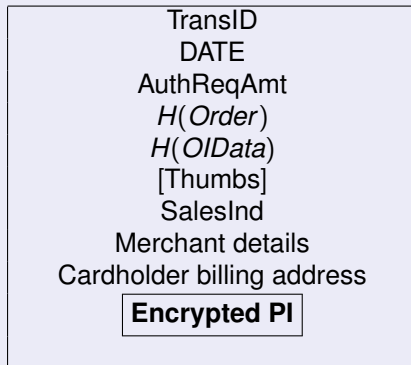
## Payment Authorization

- During the processing of an order from a cardholder, the merchant authorizes the transaction with the payment gateway.
- The payment authorization ensures that the transaction was approved by the issuer.
- This authorization guarantees that the merchant will receive payment; the merchant can therefore provide the services or goods to the customer.
- The payment authorization exchange consists of two messages: **Authorization Request (AuthReq)** and **Authorization Response (AuthRes)**.

# Payment Authorization (AuthReq/AuthRes)

AuthReq (Authorization request for the purchase)

Merchant → Payment Gateway: **AuthReq**



Signed:  $SIG_M$

Encrypt:  $KU_A$

•  $PRes = E_{KR_M}[TransID, CompletionCode, [Results], Chall\_C]$

## Payment Gateway → Merchant: **AuthRes**

The payment gateway performs the following tasks:

- Verifies all certificates
- Decrypts the digital envelope of the authorization block to obtain the symmetric key and then decrypts the authorization block
- Verifies the merchant's signature on the authorization block
- Decrypts the digital envelope of the payment block to obtain the symmetric key and then decrypts the payment block
- Verifies the dual signature on the payment block
- Verifies that the transaction ID received from the merchant matches that in the PI received (indirectly) from the customer
- Requests and receives an authorization from the issuer

## Payment Gateway → Merchant: **AuthRes**

- The acquirer (payment gateway) decrypts the message parts, verifies signatures, and checks for consistency between the purchase details sent by the merchant and those in the PI.
- If the merchant's request (AuthReqAmt) is NOT the same as the PurchaseAmt, it is checked that this difference is acceptable to policy.
- Next, the acquire obtains the authorization through the existing financial network.
- After receiving a good authorization from the card issuer (i.e., bank), an authorization response (AuthRes) is returned to the merchant with the authorization code from the issuer.

Table: AuthRes

|                        |
|------------------------|
| TransID                |
| DATE                   |
| AuthAmt                |
| AuthCode               |
| Capture token          |
| <b>CapAmt, CapCode</b> |

Signed:  $SIG_A$

Encrypt:  $KU_M$

Table: Capture Token

|                                                 |
|-------------------------------------------------|
| <b>AuthAmt, Capture Data (IDs), Token Nonce</b> |
|-------------------------------------------------|

Signed:  $SIG_A$

Encrypt:  $KU_A$  (acquirer's eyes only)

## Payment Gateway → Merchant: **AuthRes**

- Upon receiving a good authorization, a merchant can ship the goods purchased.
- A good authorization indicates that the card issuer has verified the card details.
- The good authorization indicates that the card issuer has verified the credit limit.
- Finally, the good authorization also indicates that the card issuer has given the go-ahead for the purchase.



# Capture of Payment (CapReq/CapRes)

Merchant → Payment Gateway (Acquirer): **CapReq**

Table: CapReq

|                                     |
|-------------------------------------|
| CapID (randomly globally unique ID) |
| TransID                             |
| DATE                                |
| AuthAmt                             |
| Capture token                       |

Signed:  $SIG_M$

Encrypt:  $KU_A$

Table: Capture Token

|                                                 |
|-------------------------------------------------|
| <b>AuthAmt, Capture Data (IDs), Token Nonce</b> |
|-------------------------------------------------|

Signed:  $SIG_A$

Encrypt:  $KU_A$  (acquirer's eyes only)

# Capture of Payment (CapReq/CapRes)

## Payment Gateway (Acquirer) → Merchant: **CapRes**

Table: CapRes

|                                              |
|----------------------------------------------|
| CapID                                        |
| TransID                                      |
| CapCode (Success/Failure)                    |
| CapAmt (Amount credited to merchant account) |
| Signed: $SIG_A$                              |
| Encrypt: $KU_M$                              |

- The capture response (CapRes), signed and encrypted, contains a success indication, the settled amount, and a capture code from the financial network.
- After a successful capture, the merchant has received the actual payment of money from the cardholder's purchase.

## Cardholder → Merchant: **InqReq**

- The inquiry messages allow a cardholder to verify the status of a transaction.

Table: InqReq

|                         |
|-------------------------|
| TransID                 |
| Chall_C2 (Random nonce) |
| Signed: $SIG_C$         |

# Cardholder Inquiry (InqReq/InqRes)

Merchant → Cardholder: **InqRes**

Table: InqRes

|                                         |
|-----------------------------------------|
| TransID                                 |
| Completion code (status)                |
| [Results] (authorization/capture codes) |
| Chall_C2 (Random nonce)                 |
| Signed: $S/G_M$                         |